Casey’s question — is TIME sufficient as BST’s one dimensionful input, or is m_Planck needed? ANSWER: time is sufficient AND the more native choice ($t_B = \ell_B/c$ = the commitment tick, equivalent to m_Planck via $c,\hbar$) — but ONE dimensionful input is irreducible (dimensional-analysis theorem: a dimensionless geometry can’t produce a dimensionful scale). Self-reference gives the dimensionless STRUCTURE, not the absolute scale. PAYOFF: anchoring on the tick makes the cc’s one residual DIMENSIONLESS — the cosmic age in ticks (number of commitment cycles). Bears on the #9 VEV tier-convention: native-time framing argues dimensionful-forced-in-tick-units → Derived (physics forced, units chosen), not Identified.

Keeper

2026-08-02

Is time sufficient, or is m_Planck needed? — Time is sufficient (and better); one input stays irreducible

Direct answer

Time is sufficient. m_Planck is not separately needed — a time scale (the commitment tick t_B) is fully equivalent to m_Planck and anchors the whole theory. And time is the *more natural* choice, because it is BST’s native emergent unit (the heat-semigroup parameter on D_IV^5), whereas m_Planck is an imported external mass. But you cannot get below one dimensionful input — that is a theorem of dimensional analysis, not a BST limitation.

The dimensional facts

With c and $\hbar$ exact (definitional conversion constants, not parameters), one dimensionful scale fixes everything, and it can be a time / length / mass — interconvertible: t_B —(×c)→ ℓ_B —($\times\hbar/c^2\ell$_B)→ m_Planck. BST’s actual anchor is ℓ_B (the Bergman length, from which G, m_e, v, m_Planck all derive as ℓ_B × dimensionless geometry). The tick t_B = ℓ_B/c is the identical input on the time axis. So a single time unit generates the reference length (via c) and the mass (via $\hbar$) — Casey’s intuition is correct.

Why time is the honest choice (self-reference gives structure, not scale)

Time in BST is native — the commitment-flow parameter (the tick), not a background. Anchoring on the tick anchors on the theory’s own clock. Casey’s “self-referential unit”: the commitment cycle counts itself, which forces all the *dimensionless* time-structure (every tick-ratio, every count). But a dimensionless geometry cannot produce a dimensionful number (theorem) — "one tick = how many seconds" is a fact about our meter-sticks, not the geometry. So self-reference gives the dimensionless *structure*; the absolute scale of one tick is the one irreducible input — a unit convention, not a free physical parameter (it changes no dimensionless prediction). Every theory takes exactly one such input; BST's is best phrased as the tick.

The payoff — it clarifies the cc result

Anchor on the tick, and the cosmological constant's one residual input — the cosmic age — becomes dimensionless: the age *in ticks* = the number of commitment cycles the universe has run. So: > Λ is forced except one pure number — how many ticks old the universe is (a dynamical initial-condition; the geometry doesn't force which epoch). That is the cleanest form of the residual — a dimensionless count of the theory's own clock, not a mysterious external scale. (It also matches the K1116/K1117 result: the residual is the cosmic age, Λ-independent in origin; expressed in native units it's a pure number.)

Bears on the #9 VEV tier-convention (for Casey's ruling)

If time is native, a dimensionful quantity (VEV, m_e) is forced in tick-units — its dimensionless content (v/m_Planck, m_e/m_Planck) is fully forced; only the seconds/GeV conversion is a convention. That argues such quantities are closer to Derived (physics forced, units chosen) than Identified (which implies a free parameter). Casey's question sharpens the #9 call: the native-time reading points toward Derived-with-unit-anchor-noted, not Identified. Recorded for the #9 ruling; Casey rules the convention.

Summary

- Time sufficient? Yes — equivalent to m_Planck via c, $\hbar$, and the more native anchor (the tick).
- m_Planck needed? No — it's the same input in mass-units; the tick is the same input in time-units, and it's BST-native.
- Can the input be eliminated? No — one dimensionful scale is irreducible (dimensionless geometry can't produce it); but it's a unit convention, not a physical parameter.
- Bonus: on the time axis, the cc's residual is the dimensionless cosmic age in ticks (cleanest form), and dimensionful-forced quantities lean Derived (#9).

— Keeper, 2026-08-02. Casey's TIME-vs-m_Planck question: time is sufficient + native (the commitment tick = ℓ_B/c, equivalent to m_Planck via c,$\hbar$); one dimensionful input is irreducible (theorem); self-reference gives dimensionless structure not absolute scale; the input is a unit convention not a parameter. Payoff: cc residual becomes the dimensionless age-in-ticks (number of commitment cycles). Bears on #9: native-time → dimensionful-forced quantities lean Derived not Identified. See K1116, K1117, #9, the evolution=dynamics-of-D_IV^5 note, SWPP/Koons-tick.